Conjecture 00000004277: an uncountable maximal type chain

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1 The necessary order-type clause

The conjecture asserts, in particular, that every maximal chain of the type lattice has order type $\omega + \omega^*$. This is a countable order. We construct a genuine uncountable chain in the full standard height-type lattice, extend it to a maximal chain, and prove that this maximal chain cannot be order-isomorphic to that printed order. Refuting this universal necessary clause refutes the conjunction. We make no claim that the separate number of all maximal chains differs from the continuum.

Here "types" means the classical height types of torsion-free abelian-group theory. Bunina's fixed arXiv version [1, Section 3.3, pp. 4–5] recalls their precise quotient, order and coordinate operations; she calls them *h-types* to distinguish them from complete first-order logical types. We formalize this standard combinatorial type lattice itself. We do not claim to formalize the separate classification of rank-one groups by types.

2 The full standard quotient and its order

Let $\mathcal{P}$ be the actual set of natural primes and put

$$\mathcal{C} = (\mathbb{N} \cup \{\infty\})^{\mathcal{P}}.$$

Lean uses all such characteristics, not only the special zero/infinity ones used later. The standard equivalence is

$$a \sim b \quad \Longleftrightarrow \quad \{p : a_p \neq b_p\} \text{ is finite and } (a_p = \infty \iff b_p = \infty) \text{ for every } p.$$

Thus changes are permitted only at finitely many *finite* coordinates. A single changed infinity coordinate already prevents equivalence; this is not equality modulo arbitrary finite changes.

To implement the quotient, define a preorder

$$a \preceq b \quad \Longleftrightarrow \quad (a_p = \infty \Rightarrow b_p = \infty \text{ for all } p) \quad \text{and} \quad \{p : a_p > b_p\} \text{ is finite.}$$

Lean proves reflexivity and transitivity, and proves that comparison in both directions is exactly $\sim$. Its actual `Antisymmetrization` is therefore the full type quotient. The theorem `standard_order` proves the exact standard bridge

$$[a] \leq [b] \iff \exists a' \sim a,\ b' \sim b \quad \forall p,\ a'_p \leq b'_p.$$

For the forward implication take $a'_p = \min(a_p, b_p)$ and $b' = b$. Only the finite bad set is changed, and the infinity guard ensures that all changed entries are finite. The converse follows from transitivity and the two equivalences.

3 Actual meet and join

We establish a genuine `Lattice` instance on the quotient. Its laws follow from finite unions of the bad-coordinate sets and the infinity guards. The separately verified compatibility formulas are

$$[a] \wedge [b] = [\, p \mapsto \min(a_p, b_p)\,], \qquad [a] \vee [b] = [\, p \mapsto \max(a_p, b_p)\,].$$

Consequently the partial order and the operations used in the proof are those of the standard type lattice. The proof does not replace the full lattice by an unrelated collection of subsets.

4 Real cuts give an uncountable chain

Both the primes and the rationals are countably infinite. Choose a genuine bijection $e : \mathcal{P} \longrightarrow \mathbb{Q}$. For each real number x define the full characteristic

$$c(x)_p = \begin{cases} \infty, & e(p) < x, \\ 0, & e(p) \geq x. \end{cases}$$

If $x \leq y$, then $c(x) \leq c(y)$ coordinatewise, hence $[c(x)] \leq [c(y)]$ in the actual quotient order. If $x < y$, rational density supplies q with $x < q < y$. At $p = e^{-1}(q)$ the two characteristics have values zero and infinity. Their infinity patterns differ, so they are not equivalent. Lean proves that $x \mapsto [c(x)]$ is injective and monotone.

Its actual range C is therefore a chain. If C were countable, this injection into the range subtype would make $\mathbb{R}$ countable, a contradiction. Thus C is uncountable.

5 A maximal chain with a different order type

Apply Mathlib's Zorn-lemma theorem `IsChain.exists_maxChain` to this actual chain. It produces an actual set M of height types with

$$\mathrm{IsMaxChain}(\leq, M), \qquad C \subseteq M.$$

Since C is uncountable, so is M. The printed order is encoded as the actual ordered sum $\mathbb{N} \oplus_{\mathrm{lex}} \mathbb{N}^{\mathrm{op}}$, whose underlying set is countable. Any actual order isomorphism $M \simeq_o \mathbb{N} \oplus_{\mathrm{lex}} \mathbb{N}^{\mathrm{op}}$ would therefore make M countable. Lean proves its nonexistence, and explicitly negates the universal order-type clause.

6 Verification

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. The package includes a new root build, transitive axiom audits for all 22 theorems and the three proof-bearing instances, and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed source, report and PDF. This is AI review, not human certification. Organizer acceptance is pending.

References

[1] E. Bunina, *Types in torsion free Abelian groups*, arXiv:2409.07728v1 (2024), Section 3.3, pp. 4–5. `https://arxiv.org/pdf/2409.07728v1`.